Fiscal Year:	FY 2020	**Task Last Updated:**	FY 11/30/2020
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Project Title:	Risk of Intervertebral Disc Damage after Prolonged Space Flight		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
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No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
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Flight Assignment:	NOTE: End date changed to 7/10/2020 per NSSC information (Ed., 11/12/19) NOTE: End date changed to 7/10/2019 per NSSC information (Ed., 2/11/19)		
Key Personnel Changes/Previous PI:			
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Task Description:	Our proposal is a Flight Definition Study that will use state-of-the-art imaging technologies to quantify morphology, biochemistry, metabolism, and kinematics for lumbar discs of crewmembers before and after prolonged space flight. Importantly, we will correlate these data with low back pain that spontaneously arises in space so as to establish pain and disc damage mechanisms that will serve as basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced intervertebral disc and vertebral changes (type and magnitude) and a prioritization of these changes as to their deleterious effects and risks for crewmember injury based on clinical findings. We hypothesize that spontaneous space-flight back pain and disc herniation are due to biomechanical and biological pathomechanisms. First, microgravity leads to higher than normal physiologic disc swelling and increased disc height that may stiffen the lumbar motion segment and cause abnormal segmental movement patterns. These biomechanical changes increase risk for annular rupture, vertebral endplate microfracture, and facet joint capsule strain. Second, increased disc swelling may alter nuclear matrix osmotic pressure and nutrient transport from endplate capillaries in adjacent vertebra. These biological changes adversely affect disc cell metabolism, causing pain and inducing disc matrix degradation. Our project directly addresses the Critical Path Roadmap Risks and Questions for NASA regarding disc injury (Integrated Research Plan (IRP) Gap-B4): Is damage to joint structure, intervertebral discs, or ligaments incurred during or following hypogravity exposure? The goal of this research is to characterize space-flight induced changes comprehensively in disc morphology, biochemistry, metabolism, and kinematics. These data will be correlated with measures of back pain intensity and disability. Crewmembers will be imaged pre-flight to establish baseline data and to characterize measurement repeatability. After long-term microgravity exposure (about 180 days on International Space Station-ISS), crewmembers will be studied while maintaining supine posture as soon as possible after return to 1-G in order to quantify the acute effects of prolonged space flight. Also, pre- and post-flight, they will be imaged supine and in standing upright posture at one body weight and again at 1.1 body weight so that MR images are obtained before and after axial body-weight loads. Pre-flight tests will be done within L-120 to L-60 days before the scheduled space flight. Post-flight tests will be undertaken 1-7 days and 30-60 days after landing to investigate re-adaptation to 1-G and to distinguish immediate and longer-term recoveries. Our proposed measures represent a comprehensive set of tests that evaluate exposure severity, potential injury mechanisms, and pain generator localization. Our research will aid understanding of spinal pain and deconditioning during prolonged microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We propose to use state-of-the-art, non-invasive imaging technologies to quantify morphology, biochemistry, metabolism, and kinematics for lumbar discs of crewmembers before and after prolonged space flight. Importantly, we will correlate these data with low back pain that spontaneously arises during prolonged microgravity and after re-adaptation to Earth gravity, so as to establish pathomechanisms that will serve as a basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced intervertebral disc and vertebral changes (type and magnitude) and a prioritization of these changes as to their deleterious effects and risks for crewmember injury based on clinical findings. Importantly, this research will have application to back-pain patients on Earth in general and specifically, to patients exposed to long-term bed rest or lack of mobility (spinal-cord injury patients as well as patients suffering lack of exercise, mobility, and obesity). This research also has application to abnormal spinal curvature and pain suffered by children wearing heavy backpacks to and from school.
	Our sample included 12 crew (female, n=2; male, n=10) with an average age of 51.3 years old (SD = 5.6 years). Several crewmembers had to stop inflight exercise due to pain or injury related to ARED (Advanced Resistive Exercise Device) use. Five of the 12 crew experienced a lumbar disc herniation at some point during the 1-2 month period following space flight. Disc herniation: At baseline scans during pre-flight, five of the 12 crew had disc herniations (6 segments total; n=2 at L4L5 and n=4 at L5S1). Of these six segments, three herniations remained present (and unchanged) throughout post-flight and recovery timepoints. Of the other three, two were worse at post-flight (n=1 at L4L5 and n=1 at L5S1) and one was not present at post-flight but returned at recovery with new symptoms (n=L4L5). There were two other segments that developed completely new episodes of disc herniation post-flight (n=1 at L5S1 and n=1 at L3L4). In total, there were reports of five new or progressing disc herniations following space flight affecting five crewmembers. Disc degeneration and water content: At baseline, Pfirrmann grade for 60 total lumbar spinal segments (12 crew, 5 segments per crew) ranged between grades 1 and 4 (1: 27% (16/60), 2: 33% (20/60), 3: 27% (16/60), 4: 7% (4/60)). Pfirrmann grade did not change for any level at either the post-flight or recovery timepoint. Additionally, we measured disc water content for each spinal segment and found no changes immediately following space flight for any of the levels, but there was a significant decrease of 8.5% in disc water content at recovery (66.8±7.1, 95% CI: 51.2-82.4) compared to baseline for L3L4 (73.0±7.6, 95% CI: 56.3-89.8; p=0.005). There was no association between space flight-related changes in disc degeneration or water content with risk for post-flight disc herniation. However, whole lumbar spine disc water content (discs between L1 and S1) was 25% lower at baseline in those that experienced a new or worsened disc herniation (n=5; 6.1.6±7.7, 95% CI: 52.0-71.1) compared to those who did not (n=7; 82.0±19.7, 95% CI: 63.8-100.2; p=0.027). Pfirrmann and water content moderately correlated throughout each time point (baseline: R2=0.45, p<0.001; post-flight: R2=0.36, p<0.001; recovery: R2=0.52, p<0.001). Spinal segment pathology: At baseline, 37% of lumbar spinal segments (22/60) had endplate irregularities. None of these pre-existing irregularities changed following space flight, but there was an incidence of only one new affected level. At baseline, 17% of lumbar spinal segments (10/60) had facet arthropathy. One of these cases disappeared following space flight. At baseline, 30% of lumbar spinal segments (18/60) had adjacent high intensity zones (HIZ). For two subjects, HIZ were no longer present following space flight (4/60). For three other subjects, HIZ were newly present following space flight (4/60). As for level-specific associations with risk for post-flight disc herniation, there were no consistent predictors from the pre-existing pathologies. However, three of the five levels that experienced a post-flight herniation had pre-existing facet arthropathy. Multifidus muscle size and composition: For average total CSA (cross sectional area) per level, there were significant

Task Progress:	decreases for pre- to post-flight at L3L4 (-17.1%, p=0.036) and L4L5 (-11.5%, p=0.019) and no significant differences remained between pre-flight and recovery. For average mCSA per level, there were significant decreases for pre- to post-flight at L3L4 (-21.2%, p=0.012) and L4L5 (-15.3%, p=0.007) and no significant differences remained between pre-flight and recovery. For average %muscle per level, there were significant decreases for pre- to post-flight at L3L4 (-3.8%, p=0.002) and L5S1 (-4.3%, p=0.014) and a significant difference remained between pre-flight and recovery at L3L4 (-3.3%, p=0.009). When comparing longitudinal changes in %muscle between the anteromedial and posterolateral aspects of the multifidus, there were significant changes in the anteromedial aspect with pre- to post-flight decreases in L3L4 (-5.9%, p=0.001) and L5S1(-5.8%, p=0.019) and a significant decrease remaining at L3L4 (-6.3%, p=0.001). We did not see any longitudinal changes in the posterolateral aspect of the muscle. Some baseline features for muscle were significantly different between those that experienced a new or worsened disc herniation following space flight and those that did not. For each separate lumbar spine level, %muscle was significantly lower at baseline for those that experienced a new or worsened disc herniation compared to those that did not. After comparison differences in %muscle between the anteromedial and posterolateral aspects of the multifidus, we found that baseline %muscle of the anteromedial aspect was significantly associated with later disc herniation, but %muscle of the posterolateral aspect was not. Neither totalCSA or mCSA at baseline were associated with later occurrence of disc herniation. Demographic factors like age and sex were not associated with later occurrence of disc herniation. Lumbosacral Spinal Alignment and Lumbar Disc Segment Kinematics: Standing LL did not significantly change in a consistent direction among crew between pre- (51.9°±11.8) to post-flight (52.9°±12.7; p=0.28). In addition, standing SS also did not significantly change in a consistent direction among crew between pre- (38.6°±9.1) to post-flight (39.1°±8.3; p=0.33). Maximum segmental kinematics were determined from three different motion conditions described in the methods. For maximum range of motion (°) in flexion/extension, there was a mean difference of 11.5% decreased range of motion (ROM) between pre- (10.4°±3.6) and post-flight (9.2°±3.9; p=0.02). For maximum range of motion (°) in lateral bending between left and right sides, there was a mean difference of 11.9% decreased ROM between pre- (8.4°±3.3) and post-flight (7.4°±3.4; p<0.01). For maximum translation (%) during any of the movement conditions, there was a mean difference of 13.4% decreased ROM between pre- (6.7±1.9) and post-flight (7.5±3.3; p=0.03). CONCLUSION Our findings indicate that pre-flight spinal health, including disc hydration and multifidus muscle quality are associated with risk for post-flight disc herniation. This is the first longitudinal spinal MRI study of astronauts before and after space flight. This study provides valuable knowledge toward establishing countermeasures to mitigate post-space flight chronic low back pain and disc herniation. This study also provides unique insights into mechanistic relationships between paraspinal muscle health and the development of chronic low back pain that are not observable from cross-sectional clinical studies.
Bibliography Type:	Description: (Last Updated: 08/25/2025)
Articles in Peer-reviewed Journals	Pandiarajan M, Hargens AR. "Ground-based analogs for human spaceflight." Front Physiol. 2020 Jun 23;11:716. https://doi.org/10.3389/fphys.2020.00716 ; PMID: 32655420; PMCID: PMC7324748 , Jun-2020
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Awards	Hargens AR. "Honorary Professor and Keynote Speaker at Seoul National University College of Medicine, Seoul, Korea. Invited Lectures: 'Artificial Gravity Exercise for Deep-Space Missions' and 'Orthopaedic Applications of Space Research,' December 2019." Dec-2019